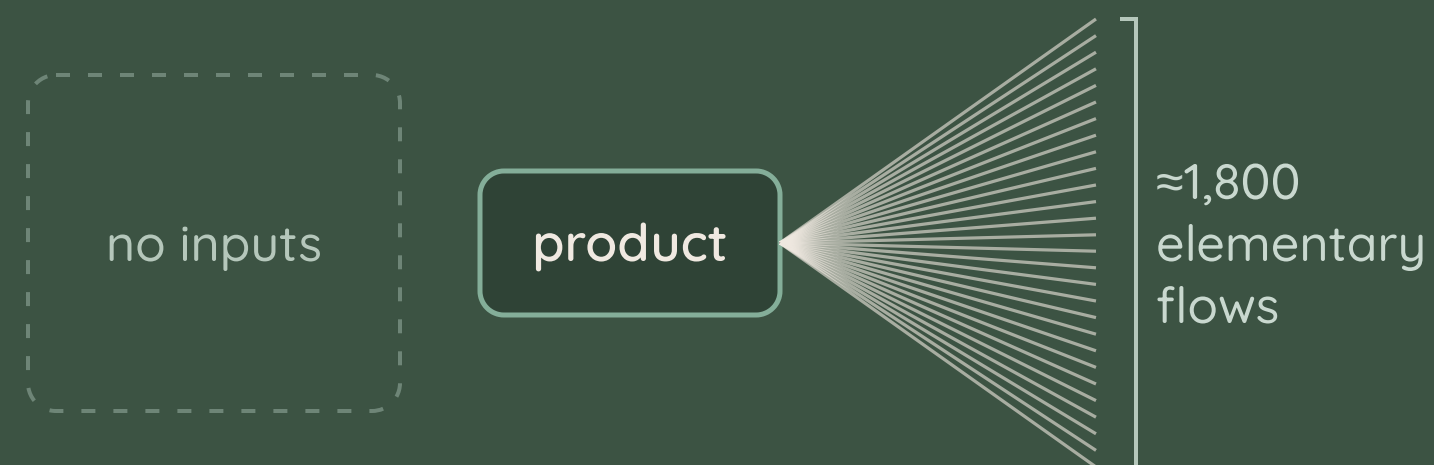


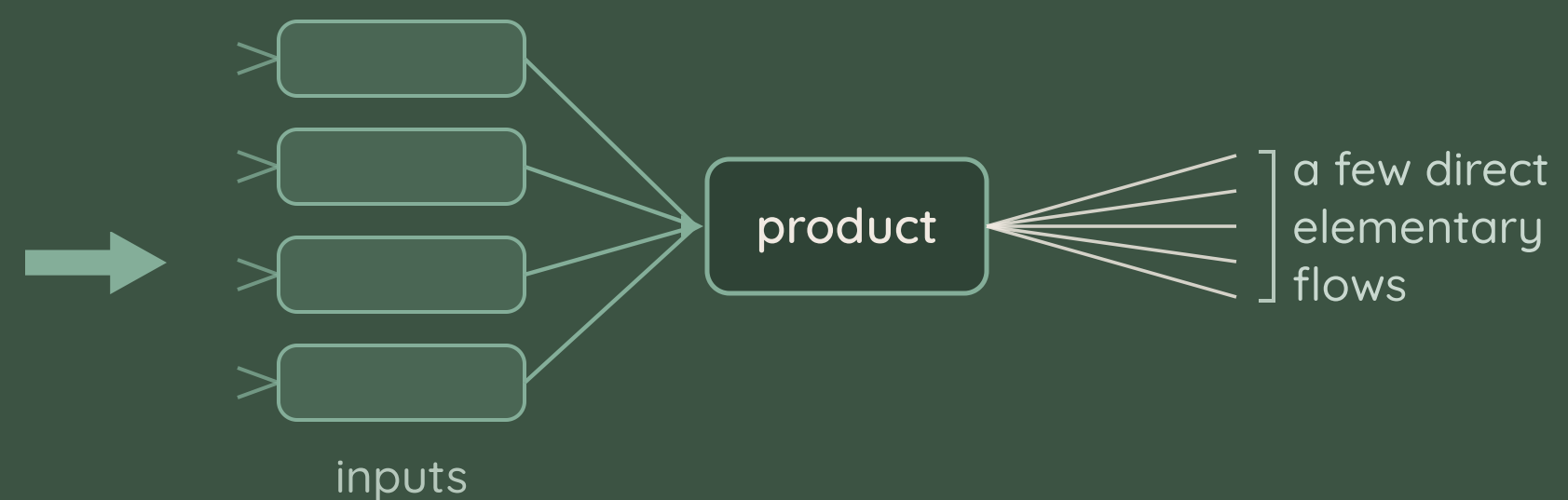
Dis-aggregating system processes

We found 138 datasets in BAFU-2026 that ship as system processes, with the sub-graph behind them collapsed away. We opened 51 of them.



system process · aggregated

Every elementary flow of the whole supply chain, attached straight to the product.



unit process · dis-aggregated

Named inputs, each carrying its own inventory, plus the emissions of this plant only.

The system processes we looked at

101

carry the ecoSpold type=2 flag

The export says so outright. ecoinvent confidential chemicals, KBOB manufacturer data, French bio-based materials.

+37

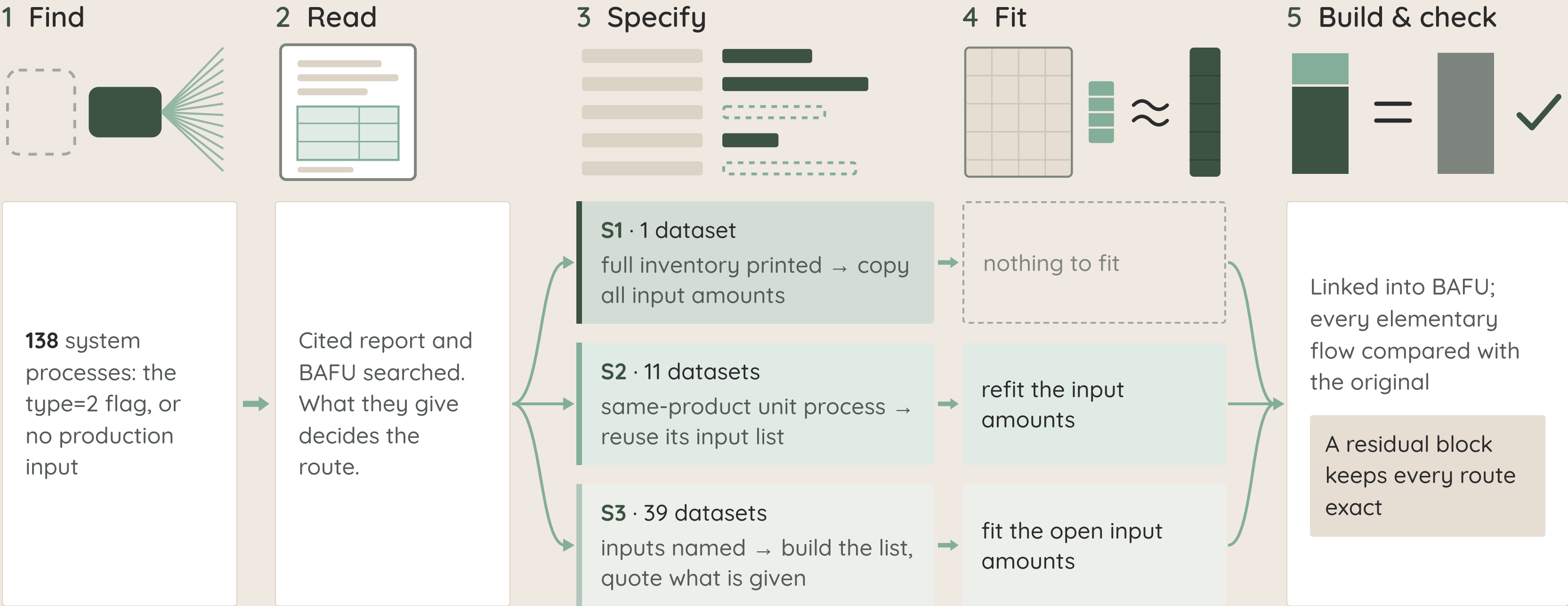
carry no flag. We found them by structure.

No production input at all — only waste-treatment services and a plant unit. Styrene, benzene, the nylons, ABS, polycarbonate, MDI/TDI: the APME era eco-profiles.

For **51** out of the 138 we found additional information in the reports that we could use for the dis-aggregation.

THE ALGORITHM AT A GLANCE

Five steps from system process to unit process



Seven inputs exact, six only as ranges

Tab. 3.14: Sachbilanz von 1kg Zement ZN/D, ab Werk

Inputs	Einheit	Menge
Klinker	%	40%-70%
Gips, mineralisch	%	< 10%
Mischgranulat (MG)	%	15%-30%
Schiefer, gebrannt	%	15%-30%
Eisen-II Sulfat	%	< 1%
Filterstaub / Kalkstein	%	< 10%
Ethylenglykol als Mahlhilfe	kg/kg	5.50E-04
Strom, Mittelspannung	kWh/kg	4.97E-02
Schweröl zur Wärmeerzeugung	MJ/kg	1.62E-01
Zementfabrik, Infrastruktur	p/kg	5.36E-11
Stahl für Mahlwerk, niedrig legiert	kg/kg	1.10E-04
Transport LKW	tkm/kg	8.37E-03
Transport Bahn	tkm/kg	4.28E-02

The dataset's own inventory (Werner 2018). Seven lines carry one number. The six shares that make up the kilogram carry only a range.

What the fit chose inside those ranges

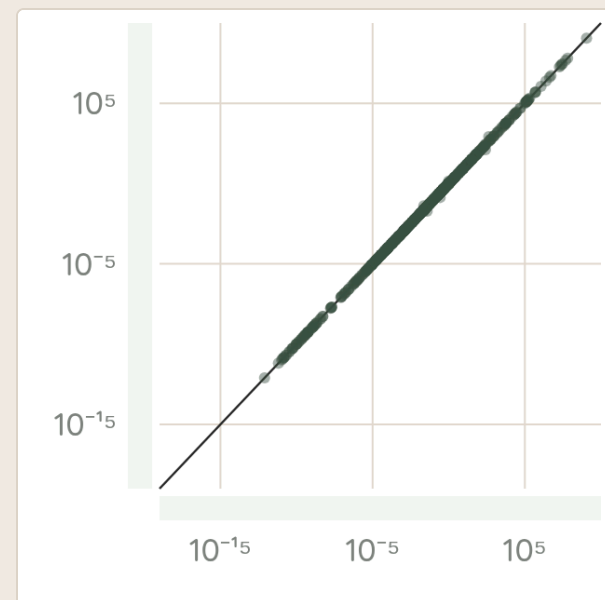
Share	Report	Fit
Klinker	40–70 %	67 %
Mischgranulat	15–30 %	18 %
Schiefer, gebrannt	15–30 %	15 % floor
Eisen-II-Sulfat	< 1 %	0 % floor
Gips	< 10 %	0 % floor
Filterstaub / Kalkstein	< 10 %	0.2 %

S3 · the input list is the report's; the composition is calibrated inside the printed ranges

RESULTS · HOW FAR CAN WE TRUST IT

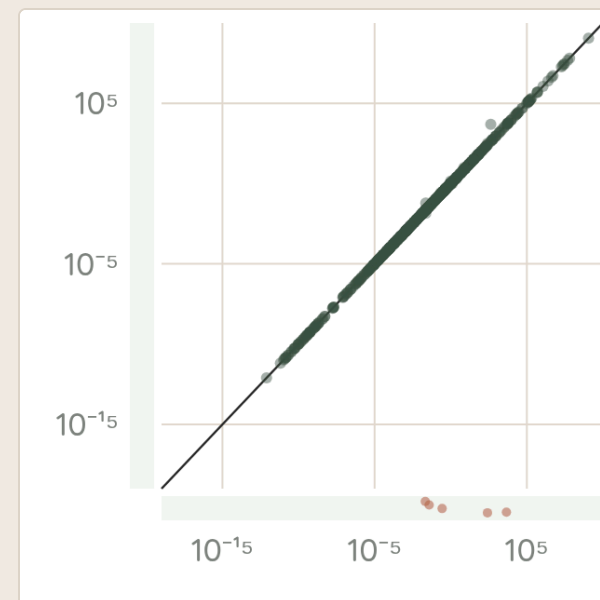
We built 100 system processes from unit processes, then dis-aggregate them

Complete input list, input amounts as ranges



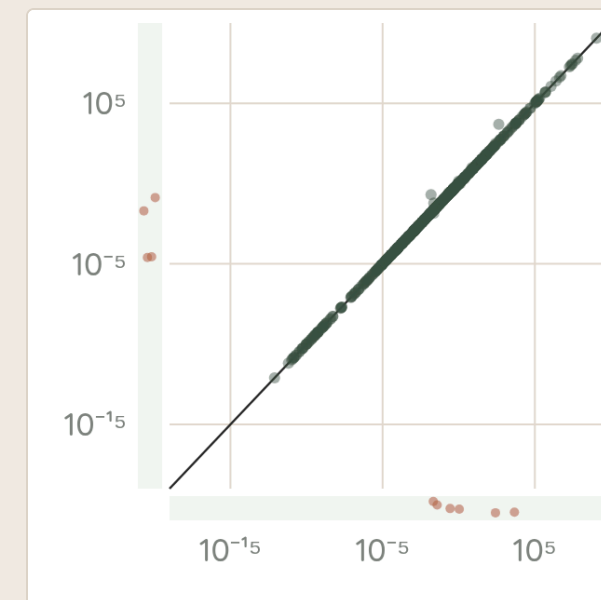
Elementary flows $\pm 10\%$: **100 %**
Input amounts $\pm 20\%$: **780 / 786**

Complete input list, no input amounts given



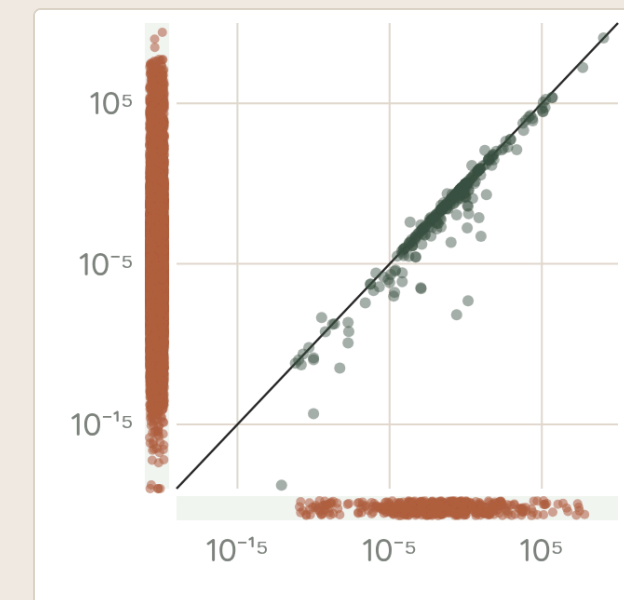
Elementary flows $\pm 10\%$: **100 %**
Input amounts $\pm 20\%$: **772 / 786**

Complete input list + 10 wrong inputs, no input amounts given



Elementary flows $\pm 10\%$: **100 %**
Input amounts $\pm 20\%$: **770 / 786**

No input list, all ~12,000 datasets on offer, no direct emissions



Elementary flows $\pm 10\%$: **100 %**
Input amounts $\pm 20\%$: **149 / 786**

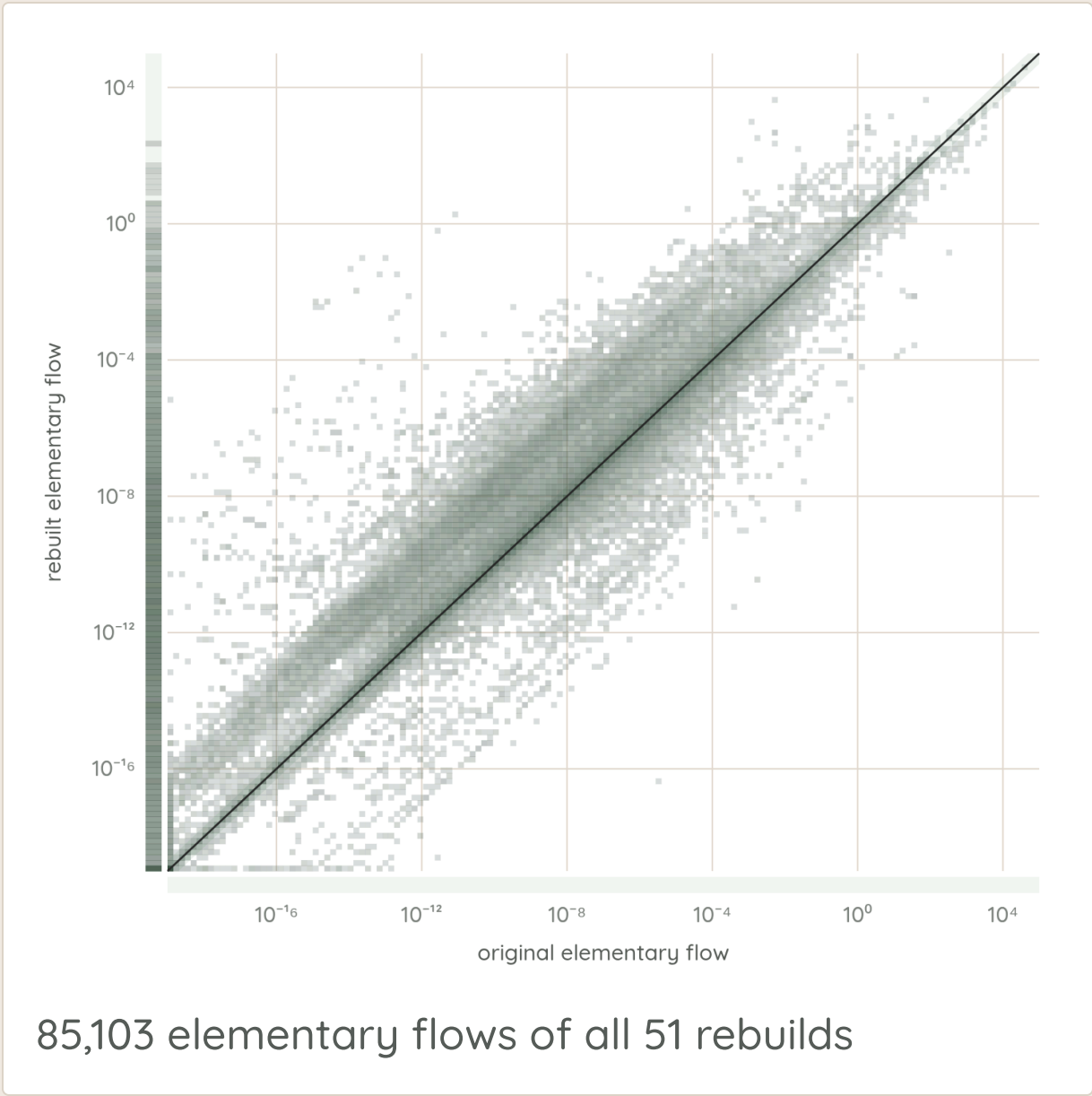
One dot per input: true input amount $\rightarrow$ against fitted $\uparrow$, on the line = recovered $\cdot$ $\bullet$ bottom strip: true input lost $\cdot$ $\bullet$ left strip: wrong input used

Given a list that contains the right inputs, the fit recovers the input amounts and drops wrong candidates.
Without a list it matches 100 % of the elementary flows with a made-up process: the inputs have to come from evidence.

The 51 system-processes dis-aggregated

n	Route
1	S1 · Transcription
11	S2 · Template transfer
39	S3 · Top-down model

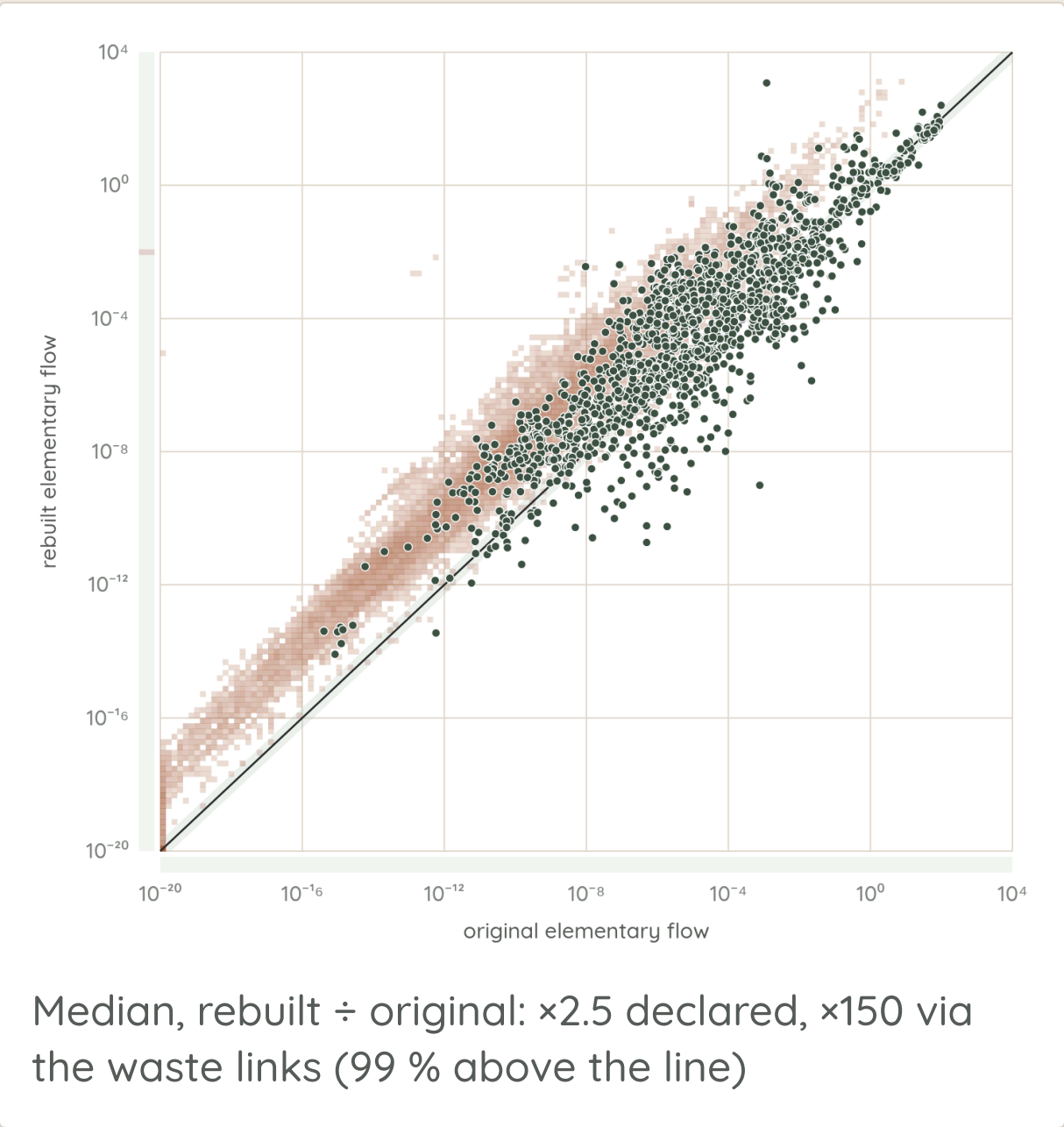
32 completely opened · **19** link to one or more system processes we did not resolve



The 12 worst matches: APME eco-profiles

Elementary flows	per dataset	rebuilt ÷ original
● declared by the eco-profile	~130	×2.5
■ only via its waste links	~1,530	×150

Epoxy resin · MDI · naphtha · polycarbonate · TDI · nylon 66 · acetone · hydrogen cyanide · benzene · polyols · GPPS · methyl methacrylate



What works, what's next

Working today

- **End-to-end pipeline:** system process + extra information (report table, free text, a same-product unit process) → a linked, traceable unit process¹
- **51 system processes rebuilt** and importable into any Brightway BAFU-2026 project²
- **Benchmarked** on 100 processes we aggregated ourselves: right input list → 772 / 786 input amounts recovered³

Next steps

- **Real-world challenges** (e.g. system processes that declare only part of their flows)
- **Little or no information** → **system experts:** peer-reviewed, transparent models for whole classes of similar products

¹ src/reverse_bafu/ · ² exports/disaggregated_bafu2026.json, scripts/import_disaggregated.py · ³ results/benchmark/flow-n100-seed7.csv

Every input has exactly one number

Tab. 3.9: Sachbilanz von 1 kg gebranntem Ölschiefer (GÖS), ab Werk (Werner 2013), angepasst gemäss Allokation aus Werner (2018).

		Zuordnung zu GÖS	
		100 %	71.30 %
Input	Einheit	pro t	pro t
Energie und Betriebsmittel			
Sprengstoff	kg	9.45E-02	6.74E-02
Diesel Abbaugerät	l	5.40E-01	3.85E-01
Stromverbrauch Brecher	kWh	1.62E+00	1.16E+00
Stromverbrauch Förderbänder	kWh	7.02E-01	5.01E-01
Stromverbrauch Zerkleinern	kWh	4.86E+00	3.47E+00
Heizöl EL, Mühle	l	2.00E-01	1.43E-01
Stromverbrauch GÖS-Produktion	kWh	4.30E+01	3.07E+01
Stromverbrauch Stromerzeugung	kWh	1.60E+01	1.14E+01
Wasserverbrauch	m³	7.00E-01	4.99E-01
Laugen/Säuren			
Salzsäure	kg	1.00E-01	7.13E-02
Natronlauge	kg	1.00E-01	7.13E-02
Chlorbleichlauge	kg	1.00E-01	7.13E-02
Chemikalien, org.	kg	8.00E-02	5.70E-02
Stromverbrauch Mahlen	kWh	3.00E+01	2.14E+01

What the page gives

The dataset’s own inventory (Werner 2013, 71.3 % economic allocation).
All 14 input lines carry one number each.

What we did

Transcribed. The only numbers we add are unit conversions: litres of diesel and heating oil to MJ, and the distances of Tab. 3.10 to tonne-kilometres.

S1 · the dataset’s own inventory, every input amount one printed number: transcription

Catch: the rebuild we count lets the fit replace the six electricity lines, 68.6 kWh/t, with 1.5 kWh/t. The draft keeps them as printed. Open decision.

Named, but never given a number

Tab. 44.3 Input and output data for the production of anthraquinone (information from Vogel (2000))

		chromic acid	oxidation with air	naphtha-lene	phthalic anhydride
Input					
water	kg	9000			
air (flow rate)	m ³ /h		2150	x	
anthracene	kg	2600	20 g/m ³		
sulphuric acid, 48%	kg	10200			
sodium dichromate, 20%	kg	23500			
naphthalene	kg			x	
xylene	kg			x	
butadiene	kg			x	
Output					
anthraquinone	kg	3000	x	x	

Energy input: In case of the chromic acid process no information about energy production is given. Therefore, it is assumed that the energy consumption is similar like in the other dataset. For the electricity consumed, an average European medium voltage mix (UCTE-mix) is used. The used steam is shown as "steam, for chemical processes, at plant (RER)" according to the description in Zah & Hischier (2007).

What the page gives

Four chemicals, for one of four routes. Below the table, electricity, steam, plant and transport are named in prose, with no amount.

What the fit did

Five inputs were left to it. It pushed four to their lower bound: steam, plant, rail and lorry are effectively zero.

S3 · the report names the inputs, the optimiser sets the input amounts

Real numbers, but not this inventory

85.6 Life cycle Inventory for titanium dioxide

This inventory is based on confidential information provided by industry sources. The input information may not be divulged. Some data can be compared with minimum and maximum values of all German production locations (UBA BAT Notes, 2001).

85.6.1 Precursor materials

Tab. 85.3 precursors for TiO2 production

Kg per kg of product	Sulfate process	Chloride process	Source
Sulphuric acid	2.4 – 3.5		(UBA BAT Notes, 2001)
Rutile, ilmenite	Ca. 2	na	Industry sources
Synthetic rutile	ng	1.06	(Chemlink.com)
Chlorine	na	0.18	(UBA BAT Notes, 2001)

What the page gives

The first sentence decides it: the inventory is confidential. The tables are minimum and maximum values across German plants, printed for comparison.

What that means

Copying them gives a plausible TiO₂ process, not this dataset. Hydrogen cyanide fails the same way: its tables are literature, the dataset is a cumulated eco-profile.

S3 · a top-down model built from published comparison data (UBA BAT notes)